

Program Schedule of International Conference
Chemical Constellation Cheminar-2026: Next Generation Materials for Energy and Environment
April 03-04, 2026

Day 1: April 03, 2026

Inaugural Ceremony: 9:30-10:15 AM
Chief Guest: Professor Mahavir Singh, Vice Chancellor, Himachal Pradesh University Shimla
Guest of Honour: Prof B K Kanaujia, Director, Dr B R Ambedkar National Institute of Technology Jalandhar
Venue: Video Conference Hall, Dr B R Ambedkar National Institute of Technology Jalandhar

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Chief Guest: Professor Mahavir Singh, Vice Chancellor, Himachal Pradesh University Shimla
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High Tea 10:15 AM-10:35 AM	
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Session 1
Session Chair: Professor N C Kothiyal, **NIT Jalandhar** **Co-chair:** Dr. Sadhika Khullar, **NIT Jalandhar**
Coordinators: Ms. Shreya Rana & Ms. Anoushka

Session 1
Session Chair: Professor N C Kothiyal, **NIT Jalandhar** **Co-chair:** Dr. Sadhika Khullar, **NIT Jalandhar**
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Time	Event	Speaker	Title
10:35-11:15 AM	Keynote Address	Professor Mahavir Singh Vice Chancellor Himachal Pradesh University Shimla	Magnetic Nanotechnology: Innovations, Challenges and Energy Applications
11:15-11:45 AM	Plenary Talk	Professor S. K. Mehta Panjab University, Chandigarh & Former Vice Chancellor, Ladakh University, India	Integrating Indian Knowledge System with Modern Green Chemistry for a Sustainable Future
11:45 AM-12:15 PM	Plenary Talk	Prof. Gurdeep Singh Former Vice Chancellor, Vinoba Bhave University, Hazaribag, India	Sustainable Mining Development with Best Management Practices

Session 2 Invited Talks									
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Hall 1
Chair: Dr Sangeeta Obrai, NIT Jalandhar
Coordinator: Ms Bindu Dhruva

Hall 2
Chair: Dr Poonam Rani, **NIT Jalandhar**
Coordinator: Ms Alka

Hall 3
Chair: Dr Manisha Devi, **NIT Jalandhar**
Coordinator: Ms Shikha Sharma

Time	Speaker	Title	Speaker	Title	Speaker	Title
12:25 -12:45 PM	Dr Jai Praksah NIT Hamirpur	Emerging Photocatalysts as Sustainable and Recyclable Nanomaterials: Multifunctional Applications from Environment to Health	Dr Jaraymulu Kolleboina IIT Jammu	Single-Atom Porous Catalysts for Cross-Coupling Chemistry	Professor Deepak Pathania Central University Jammu	Biomass-based Sustainable Fluorescent Sensors for Monitoring of Wastewater Contaminants
12:45 -1:05 PM	Dr Vipin K Saini Doon University, Dehradun	Designing Functional Clay-Based Porous Materials for Sustainable Energy and Environmental Applications	Dr Nitin Singhal NABI Mohali	DNA Origami-Based Aptamer Nanosensor for Rapid Detection of Food-Borne Pathogens Using Multivalent Target Recognition	Dr Himanshu Rani SLIET Longowal	RSM Optimized Supercritical Extraction of Inulin from <i>Helianthus tuberosus</i> L. and subsequent modification by pyridine moiety for Antioxidant Applications

Session 3 Oral Presentations						
1:05-1:20 PM	Ankita Sharma NIT Hamirpur	3-amino coumarin Schiff base as highly selective and sensitive fluorescent probe of Cu ²⁺ : Spectroscopic, computational, cell imaging studies and logic gate construction	Sonali Rana Himachal Pradesh University Shimla	From Pine Needles to High-Performance Adsorbent: Epichlorohydrin-Linked PANI Composite for Cationic Dyes	Navdeep NIT Jalandhar	MXene-Based Electrocatalysts for Electrochemical Water Splitting: From Synthesis to Structure–Performance Relationships
1:20 -1:35 PM	Neha Chauhan NIT Jalandhar	Enzyme immobilization on plant-mediated nanoparticles for textile effluents treatment	Rohit NIT Jalandhar	Biowaste Derived Biosorbent for Water Uptake From Petroleum Fractions-Water Emulsions	Payal Malik SLIET Longowal	C(sp ³)-H Bond Oxidation of Toluene using Mn(III)-Schiff Base Complex under Solvent-free Conditions
Lunch Break (1:35-2:30 PM), Venue: The Snackers						
Session 4 Plenary Talks Session Chair: Dr Pragya Ranjan Raut, NIT Jalandhar Coordinators: Ms. Himani Sharma						
Time	Event		Speaker		Title	
2:30-3:00 PM	Plenary Talk		Professor Narinder Singh, IIT Ropar		Nanomaterials for Water Purification and Quality Monitoring	
3:00-3:30 PM	Plenary Talk		Professor Sanjay Mandal, IISER Mohali		Tuneable Three-Component Metal-Organic Frameworks Consisting of Spanning Linkers with Six N/O-donor Atoms	
3:30-4:00 PM	Plenary Talk		Professor M R Maurya, IIT Roorkee		Molybdenum and vanadium Complexes catalyse One-Pot-Multicomponent Reactions Producing Biologically Active Molecules	
4:00-4:30 PM	Plenary Talk		Professor Rajendra Srivatava, IIT Ropar		Heterogeneous Catalyst Design for Reductive transformations of Biomass and Plastic to Chemicals and Fuels	
Session 5 Invited Talks						
Hall 1 Chair: Dr Uma Shanker, NIT Jalandhar Coordinator: Mr Abhinav Gupta			Hall 2 Chair: Dr Rakesh Kumar, NIT Jalandhar Coordinator: Ms Shreya Rana		Hall 3 Chair: Dr Sumer Singh Meena, NIT Jalandhar Coordinator: Ms Anjali	
Time	Speaker	Title	Speaker	Title	Speaker	Title

4:40 -5:00 PM	Professor S S Chimni Guru Nanak Dev University Amritsar	Enantioselective new bond formation: A nature inspired approach	Dr Rahul Saxena Ex. President and CSO, Navin Fluorine International Ltd Mumbai	Fluorine at the Frontier: Transforming Medicine, Crops, and Advanced Materials	Dr Amardeep Singh Indian Oil Corporation Ltd.	Clarifying Agents in Polypropylene: Growth Strategies & Future Outlook
5:00 -5:20 PM	Dr Kiran Singh NIT Jalandhar	Multifunctional properties of materials	Manash Pratim Sarmah Syngene International Limited Bangalore	(i) Iodine-Catalyzed Vinylogous Friedel–Crafts Alkylation of Indoles: Facile Access to Versatile β -(3-Indolyl)carbonyl Frameworks & (ii) Hypodiboric Acid–Sodium Carbonate Mediated Reductive Cyclization of Nitro(Hetero)Arenes: Facile Syntheses of certain Heterocycles	Professor Pravin Ingole IIT Delhi	Integrated approach for CO ₂ capture and electrochemical conversion: a step forward to net zero goal

Session 6
Oral Presentations

5:20 -5:35 PM	Reeta NIT Hamirpur	Surface-Engineered Mesoporous Silica Hybrids Anchored with Oxo-Vanadiumflavonoid metal Complexes as for Targeted Health Care in Diabetes Management	Sanchit Kalra Panjab University Chandigarh	Synergistic Engineering of Cationic Polymer-Ag Catalyst Matrix with Enhanced Electrochemical Generation of Reactive Oxygen Species for Energy-Efficient Pathogen Inactivation at Lab-Scale and in Municipal Effluent	Astha DAV University Jalandhar Online	In ³⁺ -Doped LaFeO ₃ Nanoparticles for Binary Dye Degradation in Water :Environmental Remediation
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Tea Break 5:35-6:00 PM

Session 7
Poster Presentations
Hall 4

(5:45-7:00 PM)

Session Chairs: Dr Uma Shanker & Dr Sadhika Khullar

Coordinators: Mr Janak & Ms Saksham

Maximum time for each poster presentation will be 5 minutes

Dinner: 7:30 PM onwards, Venue: The Snackers

Day 2: April 04, 2026

Session 8

Chair: Professor B S Kaith, **NIT Jalandhar**

Co-chair: Dr Rakesh Kumar, **NIT Jalandhar**

Coordinator: Ms. Shubhanjali Choudhary

Time	Event	Speaker	Title
9:00-9:45 AM	Keynote Address	Professor Ajay Mishra University of Western Cape, South Africa	Development of the Advanced Materials using Organic-Inorganic Hybrid Materials
9:45-10:15 AM	Plenary Talk	Prof. Debaprasad Mandal IIT Ropar	Beyond Li-ion: High Energy Aqueous Metal-sulfur and Metal-air Batteries
10:15-10:45 AM	Plenary Talk	Professor Shivani Mishra Academy of Nanotechnology and Wastewater Innovations, South Africa (Online mode)	Sol Gel Technology: Nanoscale methodology for industrial and environmental applications
10:45-11:15 AM	Plenary Talk	Professor Dalila Hammiche University of Bejaia, Algeria (Online mode)	Composite materials used in the construction sector

Tea Break
(11:15-11:35 AM)

Session 9 Invited Talks

Hall 1 Chair: Dr Kiran Singh, NIT Jalandhar Coordinator: Mr. Janak			Hall 2 Chair: Dr Anee Mohanty, NIT Jalandhar Coordinator: Ms. Shikha Sharma		Hall 3 Chair: Dr Rakesh Kumar, NIT Jalandhar Coordinator: Ms. Shubhanjali Choudhary	
Time	Speaker	Title	Speaker	Title	Speaker	Title
11:40 AM-12:00 PM	Prof. Venkata Krishnan IIT Mandi	Non-Noble Metal based Sustainable Heterogeneous Catalysts for Chemical Recycling and Upcycling of Plastic Waste	Dr Geetharani K Indian Institute of Science Bangalore	Borylation Reactions using Main Group Compounds and Metal Catalysts	Dr Amit Kumar Pohang University of Science and Technology, Republic of Korea (Online mode)	NanoChemoBionics: Engineering Chemical Continuity between Life and Matter
12:00-12:20 PM	Dr Manviri Rani MNIT Jaipur	Phyto-Engineered rGO–Spinel and Polymer–Derived Metal Oxide Interfaces for Solar–Driven Molecular Abatement of Environmental Pollutants	Prof S K Kansal Punjab University Chandigarh	Next-Generation Photocatalysts for Simultaneous Water Purification and Green Hydrogen Production	Dr Siddharth Chopra CSIR-CDRI Lukhnow (Online mode)	Drugs for Bad Bugs : An update

Session 10 Oral Presentations

12:20 -12:35 PM	Dr Amit Kumar Sharma Chandigarh University Mohali	Natural Polymer based hydrogel nanocomposites: A sustainable platform for wastewater treatment	Shikha Panjab University Chandigarh	Natural Gum-Based Hydrogel from <i>Nyssa sylvatica</i> : A Promising Platform for Cationic Dye Remediation	Hritankhi Tripathy NIT Rourkela (Online mode)	Dual-Defect Engineered Photocatalytic Conducting Polymer Foamgel for Antibiotic Degradation and Simultaneous Energy Harvesting
12:35 -12:50 PM	Ankita	Lovely Professional University Phagwara Bio-Conjugated Conducting Polymer for Optical-Electrochemical Application	Sapna Sharma Himachal Pradesh University Shimla	Maize Starch-derived NIPU Hydrogel for pH-Enzyme Synergistic Intestinal Drug Delivery	Swati, Maharaja Ranjit Singh Punjab Technical University, Dabwali (Online mode)	Chalcone-based Triazoles as Potent Anticancer Agents
12:50 -1:05 PM	Pawan Kumar NIT Srinagar	Facile Synthesis of Blue-Emissive Carbon Dots for the Detection of Fe ²⁺ and Fe ³⁺ Ions via a Turn-off Mechanism	Urvika DAV University Jalandhar	A Schiff Base Ligand as a Visual Probe for Dual Cation–Anion Detection and Protein Interaction Studies	Ashish Kumar DAV University Jalandhar (Online mode)	Eliminating Mixed Cationic and Anionic Dye Pollutants via Nickel-Doped Calcium Ferrite Photo catalyst Calcined at 800°C

**Lunch Break,
(1:05-2:00 PM)
Venue: The Snackers**

Session 11

Invited Talks

Hall 1 Chair: Dr Sadhika Khullar , NIT Jalandhar Coordinator: Ms. Anjali			Hall 2 Chair: Dr Poonam Rani, NIT Jalandhar Coordinator: Ms. Simran		Hall 3 Chair: Dr Manisha Devi, NIT Jalandhar Coordinator: Mr. Abhinav Gupta	
Time	Speaker	Title	Speaker	Title	Speaker	Title
2:00-2:20 PM	Dr Manoj Kumar Nayak CSIO Chandigarh	Metal–Organic Framework Platforms for Sensitive Detection of Foodborne and Environmental Contaminants	Professor R. K. Dutta IIT Roorkee	Nanoscale Materials for Sensing, Environmental Remediation, Energy Generation And Storage	Dr Shubhankar Bose Jain University Bangalore	Efficient Synthesis of Multisubstituted Alkenyl boronates <i>via</i> Difunctionalization of Alkynes: From Earth-Abundant Catalyst to Catalyst-Free Process
2:20-2:40 PM	Dr Aman Bhalla Panjab University, Chandigarh	Synthesis, Characterization and Biological Evaluation of Functionalized β -Lactam Scaffolds	Professor Ashok Kumar Sharma Deenbandhu Chhotu Ram University, Haryana	Electrode Materials for Advanced Energy Storage and Conversion Devices	Dr Sunil Kumar Himachal Pradesh University Shimla	Zinc Oxide Based Electrochemical Nanosensors

Session 12

Oral Presentations

2:40-2:55 PM	Ritika Soni Lovely Professional University Phagwara	Advanced Ti_3C_2 -Integrated $\text{BFO/Fe-g-C}_3\text{N}_4$ Nanostructures for Efficient Photocatalytic Degradation of Organic Dyes	Rutu Parna NIT Jalandhar	Fabrication of Hydrophobic $\text{TiO}_2\text{-SiO}_2\text{-PDMS}$ Composite Coatings via Dip-Coating	Ashish Kumar Soni Lovely Professional University Phagwara Online	Decoding Thermoluminescence through First-Principles Modelling: Defect Chemistry and Emission Mechanisms in $\text{CaSO}_4\text{:Dy}$, LiF -Based Phosphors and Optically Stimulated Luminescence (OSL) dosimeters
2:55 -3:10 PM	Isham Preet Kaur NIT Jalandhar	Bioactive Compounds from Extremophiles: A Study on Stability and Functional Properties	Shubhanjali Choudhary NIT Jalandhar	Aryne-mediated one-pot green synthesis of benzofuroquinolines and benzofuopyridines via tandem cycloaddition-cyclization of haloquinoline- and halopyridine <i>N</i> -oxides	Pritika Khullar Lovely Professional University Phagwara Online	An integrative study of semi- synthetic derivatives of Jatamansone from <i>Nardostachys Jatamansi</i> -medicinal plant
3:10- 3:25 PM	Pallavi Sharma NIT Jalandhar	Functional Modification of Larch Wood Derived Arabinogalactan for Enhanced Biomedical Applications	Spriha NIT Jalandhar	Bioprospecting Extremophiles for Functional Bioactive Compounds	Vinita Khandegar GGSIU New Delhi Online	Ni-SS Electrode for Energy and Environmental Applications
3:25-3:40 PM	Janak NIT Jalandhar	Nickel-Based Coordination Polymers as Bifunctional Materials for Oxygen Evolution Reactions and Supercapacitor applications	Shreya Rana NIT Jalandhar	Visible light photo-redox catalysis for the C-1 arylation of β -Carboline <i>N</i> -oxides using arene diazonium salts	Aicha DEHANE University of Bejaia & NIT Jalandhar Online	Green-Synthesized Copper Oxide/Chitosan Smart Hydrogel Nanocomposites as Eco-Friendly Slow-Release Urea Fertilizers in Tomato Production

Valedictory Ceremony

(3:45 PM onwards)

Venue: NKN Hall, Dr B R Ambedkar National Institute of Technology Jalandhar

Chairs: Dr Uma Shanker and Dr Sadhika Khullar, NIT Jalandhar